19 (a) Write 168 as a product of its prime factors.
You must show your working.

2	168
2	84
2	42
3	21
7	7
	1

$168 = 2 \times 2 \times 2 \times 3 \times 7$

$= 2^3 \times 3 \times 7$

1 mark
Continual division

1 mark
All prime factors

1 mark
Product notation

(3)

(b) Find the highest common factor (HCF) of 168 and 180

$180 = 2 \times 90 = 2 \times 2 \times 45$

$= 2 \times 2 \times 3 \times 3 \times 5 = 2^2 \times 3^2 \times 5.$

Common primes with 168: $2 \times 2 \times 3.$

$HCF = 12.$

1 mark
Prime factorisation 180

1 mark
Common product 12

(2)

(Total for Question 19 is 5 marks)